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Hanger assembly with alignment/anti-rotation system

Abstract

A hanger assembly has at least one elongated channel disposed on the outer surface of the hanger assembly. An alignment/anti-rotation tab member is moveably disposed within each elongated channel and can selectively alternate between a first retracted configuration and a second extended configuration. In the second extended configuration, each alignment/anti-rotation tab member can be received within a mating channel along the inner surface of the central through bore of the wellhead. Each extended alignment/anti-rotation tab member can engage against a mating recess or channel in order to align and orient the hanger assembly within the central through bore of the wellhead (permitting orientation and alignment of ports in the hanger assembly and the surrounding wellhead), while also preventing rotation of the hanger assembly relative to the wellhead.

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Background/Summary

CROSS REFERENCES TO RELATED APPLICATION (1) THIS APPLICATION CLAIMS PRIORITY OF U.S. PROVISIONAL PATENT APPLICATION Ser. No. 63/401,777, FILED Aug. 29, 2022, INCORPORATED BY REFERENCE HEREIN.

STATEMENTS AS TO THE RIGHTS TO THE INVENTION MADE UNDER FEDERALLY SPONSORED RESEARCH AND DEVELOPMENT

(1) NONE

BACKGROUND OF THE INVENTION

1. Field of the Invention

(2) The present invention pertains to a hanger assembly for use in oil and gas wells. More particularly, the present invention pertains to a hanger assembly having at least one alignment and/or anti-rotation member. More particularly still, the present invention pertains to a hanger assembly having at least one alignment or anti-rotation member that selectively provides specific orientation, and prevents rotation, of said hanger assembly relative to a mating wellhead.

2. Description of Related Art

(3) During the drilling of oil and/or gas wells, wellhead systems are usually installed at or near the earth's surface. Such wellhead systems are typically assembled as the drilling process proceeds. Generally, wellhead systems typically comprise a starting casing head, at least one intermediate casing wellhead and, ultimately, a tubing wellhead.

(4) A casing hanger, which is a device attached to the uppermost casing joint in a well, is typically received or “landed” within a casing wellhead to support a casing string. Similarly, a tubing hanger is a device attached to the uppermost joint of production tubing in a well; the tubing hanger is

received or “landed” within a tubing wellhead to support the tubing string.

(5) Both hangers and mating wellheads (casing heads or tubing heads) frequently incorporate some type of sealing system to ensure that the central bore of a pipe string is hydraulically isolated from its annulus. Further, hangers often include at least one port to permit the communication of hydraulic fluid, electrical signal(s), chemical injection and/or other substances into said hanger via said port(s). Similarly, mating wellheads (casing heads or tubing heads) also frequently include at least one port to allow for many of the same functions and/or benefits as the port(s) in said hanger. Ideally, the port(s) in a hanger should be aligned with the port(s) in a surrounding wellhead.

(6) Conventional hangers do not provide a means for orienting and aligning said port(s) in said hangers with port(s) in a surrounding wellhead in order to allow unrestricted flow through said aligned port(s). Additionally, conventional hangers also do provide a means for preventing the hanger from rotating within a wellhead, which can cause said hanger port(s) to become misaligned with wellhead port(s).

(7) The misalignment of port(s) between the hanger and wellhead can negatively impact functional operation of the equipment. Further, such misalignment can cause the hanger and wellhead to experience erosion from fluid flow, which can cause damage that can necessitate the costly removal, repair and/or replacement of the hanger and/or wellhead.

(8) Thus, there is a need for an efficient and effective system for aligning the port(s) in a hanger with port(s) in a surrounding wellhead. The system should further prevent a hanger from rotating within a wellhead and thereby causing said hanger port(s) to become misaligned with port(s) in said wellhead. Further, the system should be cost-effective and easy to manufacture and operate.

SUMMARY OF THE INVENTION

(9) In a preferred embodiment, the present invention comprises a hanger assembly for use within a mating wellhead. Said hanger assembly includes a mechanism for aligning port(s) in said hanger with port(s) in said surrounding wellhead. The present invention further prevents said hanger from rotating within a surrounding wellhead which can cause said hanger port(s) to become misaligned with port(s) in said wellhead.

(10) A hanger assembly is configured to be received within the central through bore of a wellhead. Said hanger assembly can be disposed at the upper end of a string of casing, tubing or other pipe which can be “hung” or “landed” within said wellhead. At least one downwardly-facing shoulder is disposed on the outer surface of said hanger assembly and can cooperate with an opposing upwardly-facing shoulder disposed on the inner surface of said central through bore of said wellhead. Said opposing shoulders cooperate to support said hanger assembly and any attached pipe string.

(11) In a preferred embodiment, the hanger assembly and wellhead of the present invention further comprise at least one alignment/anti-rotation tab system. Said hanger assembly has a central through-bore, and at least one elongated channel is disposed on the outer surface of said hanger assembly. Said elongated channel is oriented substantially parallel to the longitudinal axis of said central through bore. An alignment/anti-rotation tab member is moveably disposed within each such channel. Said alignment/anti-rotation tab member can selectively alternate between a first retracted configuration, and a second extended configuration.

(12) When in said extended configuration, said alignment/anti-rotation tab member can be received within a mating recess or channel disposed along the inner surface of the central through bore of said wellhead. Said extended alignment/anti-rotation tab member can engage against said mating recess or channel in order to align said hanger assembly within said central through bore of said wellhead. Further, said extended alignment/anti-rotation tab member can engage against said mating recess or channel to prevent relative rotation of said hanger assembly within said central through bore of said wellhead.

Description

BRIEF DESCRIPTION OF DRAWINGS/FIGURES

(1) The foregoing summary, as well as any detailed description of the preferred embodiments, is better understood when read in conjunction with the drawings and figures contained herein. For the purpose of illustrating the invention, the drawings and figures show certain preferred embodiments. It is understood, however, that the invention is not limited to the specific methods and devices disclosed in such drawings or figures.

(2) Further, the drawings constitute a part of this specification and include exemplary embodiments of the technology. It is to be understood that in some instances, various aspects of the invention may be shown exaggerated or enlarged to facilitate an understanding of the invention. Therefore, the drawings may not be to scale.

(3) FIG. 1 depicts a side perspective view of a hanger assembly comprising an alignment/anti-rotation member and mating wellhead assembly of the present invention.

(4) FIG. 2 depicts a side perspective cut-away view of a hanger assembly having an alignment/anti-rotation member and mating wellhead assembly of the present invention.

(5) FIG. 3 depicts a side perspective cut-away and partially exploded view of a hanger assembly of the present invention having a plurality of alignment/anti-rotation tab members.

(6) FIG. 4 depicts a side sectional view of a hanger assembly of the present invention having a plurality of alignment/anti-rotation tab members in a retracted configuration (such as, for example, for shipment).

(7) FIG. 5 depicts a side perspective view of a hanger assembly of the present invention having an alignment/anti-rotation tab member in a retracted configuration (such as, for example, for shipment).

(8) FIG. 6 depicts a side perspective and cut away view of a hanger assembly of the present invention having a plurality of alignment/anti-rotation tab members in an extended configuration.

(9) FIG. 7 depicts a side perspective view of a hanger assembly of the present invention having an alignment/anti-rotation tab member in an extended configuration.

(10) FIG. 8 depicts a side sectional view of a hanger assembly of the present invention within a wellhead with alignment/anti-rotation tab members in a retracted configuration.

(11) FIG. 9 depicts a side sectional and detailed view of an alignment/anti-rotation tab member in a retracted configuration.

(12) FIG. 10 depicts a side sectional view of a hanger assembly of the present invention within a wellhead with alignment/anti-rotation tab members in an extended configuration.

(13) FIG. 11 depicts a side sectional and detailed view of an alignment/anti-rotation tab member in a retracted configuration.

(14) FIG. 12 depicts a side perspective cut-away view of a hanger assembly of the present invention landed within a mating wellhead.

DETAILED DESCRIPTION OF A PREFERRED EMBODIMENT OF THE INVENTION

(15) Before describing various embodiments of the present disclosure in further detail by way of exemplary description, examples, and results, it is to be understood that the apparatus and methods of the present disclosure are not limited in application to the details of specific embodiments and examples as set forth in the following description. The description provided herein is intended for purposes of illustration only and is not intended to be construed in a limiting sense. As such, the language used herein is intended to be given the broadest possible scope and meaning, and the embodiments and examples are meant to be exemplary, not exhaustive. Also, it is to be understood that the phraseology and terminology employed herein is for the purpose of description only and should not be regarded as limiting unless otherwise indicated as so. Moreover, in the following detailed description, numerous specific details are set forth in order to provide a more thorough

understanding of the present disclosure.

(16) It will be apparent to a person having ordinary skill in the art that the present disclosure may be practiced without these specific details. In other instances, features which are well known to persons of ordinary skill in the art have not been described in detail to avoid unnecessary complication of the description. It is intended that all alternatives, substitutions, modifications, and equivalents apparent to those having ordinary skill in the art are included within the scope of the present disclosure. Thus, while the apparatus and methods of the present disclosure have been described in terms of particular embodiments, it will be apparent to those of skill in the art that variations may be applied to the apparatus and methods and the steps or in the sequence of steps of the methods described herein without departing from the concept, spirit, and scope of the inventive concepts. Referring to the drawings, like numerals indicate like or corresponding parts throughout the several views.

(17) FIG. 1 depicts a side perspective view of a wellhead **110** and hanger assembly **10** of the present invention. Said hanger assembly **10** and mating wellhead **110** are equipped with an alignment/anti-rotation system of the present invention. Hanger assembly **10** has central through bore **20**, while wellhead assembly **110** has central through bore **120**. Although not depicted in FIG. 1, it is to be observed that hanger assembly **10** can be attached at the upper extent of a length of pipe, such as a casing string or other tubular string, being run in a well. Said hanger assembly **10**, with a length of pipe attached, can be received or “landed” within central through bore **120** of wellhead **110** in order to support said pipe string.

(18) In the embodiment depicted in FIG. 1, hanger assembly **10** further comprises alignment/anti-rotation tab member **40** disposed on the outer surface of said hanger assembly **10**. Additionally, said hanger assembly **10** further generally comprises landing shoulder **15** which engages with an opposing shoulder (not visible in FIG. 1) disposed on the inner surface of central through bore **120** of wellhead **110**. A plurality of transverse ports **30** extend through hanger assembly **10** from the external surface of said hanger assembly to central through bore **20**; when said hanger assembly **10** is fully received and landed within central through bore **120** of wellhead **110**, said transverse ports **30** can be aligned with transverse ports **130** extending through wellhead **110**.

(19) A plurality of grooves **16** for receiving sealing members, such as O-rings or other elastomeric sealing elements, are disposed along the outer surface of said hanger assembly **10**. When said hanger assembly **10** is fully received and landed within central through bore **120** of wellhead **110**, said O-rings or other sealing members disposed in grooves **16** can engage against the inner surface of said central through bore **120** of wellhead **110** to form a fluid pressure seal between said hanger assembly **10** and said wellhead **110**.

(20) FIG. 2 depicts a side perspective and partial cut-away view of a wellhead **110** and hanger assembly **10** of the present invention. Hanger assembly **10** has central through bore **20**, while wellhead assembly **110** has central through bore **120**. Hanger assembly **10** has downwardly-facing landing shoulder **15** that can engage against opposing upwardly-facing shoulder **115** which is disposed on the inner surface of central through bore **120** of wellhead **110**. Said landing shoulder **15** and opposing shoulder **115** cooperate to support hanger assembly **10** and any attached pipe string when said hanger assembly **10** is landed within central through bore **120** of wellhead **110**.

(21) A plurality of transverse ports **30** extend through hanger assembly **10** from the external surface of said hanger assembly to central through bore **20**. A plurality of transverse ports **130** also extend through wellhead **110** from the external surface of said wellhead **110** to said central through bore **120**. Said ports **30** of hanger **10** can be aligned with transverse ports **130** extending through wellhead **110** when hanger assembly **10** is fully received and landed (and properly aligned and oriented) within central through bore **120** of wellhead **110**. Grooves **16** are disposed along the outer surface of hanger assembly **10** for receiving O-rings or other elastomeric sealing members. Said sealing members disposed within grooves **16** can engage against the inner surface of said central through bore **120** of wellhead **110** to form a fluid pressure seal between said hanger assembly **10**

and said wellhead **110** when said hanger assembly **10** is landed within wellhead **110**.

(22) Alignment/anti-rotation tab members **40** are disposed on the outer surface of said hanger assembly **10**. Slots **140** are disposed on the inner surface of central through bore **120** of wellhead **110**. When hanger assembly **10** is fully received and landed within central through bore **120** of wellhead **110**, alignment/anti-rotation tab members **40** of said hanger assembly **10** are generally aligned with said slots **140** formed on the inner surface of central through bore **120** of wellhead **110**.

(23) FIG. 3 depicts a side perspective cut-away and partially exploded view of a hanger assembly **10** of the present invention, while FIG. 4 depicts a side sectional view of said hanger assembly **10**. Referring to FIGS. 3 and 4, hanger assembly **10** has central through bore **20**. A plurality of transverse ports **30** extend through hanger assembly **10** from the external surface of said hanger assembly **10** to central through bore **20**. Grooves **16** are disposed along the outer surface of hanger assembly **10** for receiving O-rings or other elastomeric sealing members.

(24) In a preferred embodiment, elongated recesses **70** are disposed along the exterior surface of hanger assembly **10** and are generally oriented parallel to the longitudinal axis of central through bore **20**. Alignment/anti-rotation tab members **40** are moveably disposed within said elongated recesses **70**. Alignment/anti-rotation tab members **40** each have an elongated slot **42** and transverse bore **41**. Alignment bolts **60**, each having head **61**, and alignment bolts **62**, each having head **63**, are disposed through said elongated slot **42** of each alignment/anti-rotation tab member **40**. Further, said alignment bolts **60** and **61** can be partially received within threaded bores **71** in elongated recesses **70** of hanger assembly **10**. A set screw **50** is received within each transverse bore **41** of each alignment/anti-rotation tab member **40**.

(25) When said set screws **50** are sufficiently loosened, it is to be observed that said alignment/anti-rotation tab members **40** can slidably move within said elongated recesses **70**; said movement can be in the direction of the longitudinal axis of each of said elongated recess **70**. In the embodiment depicted in FIG. 4, alignment/anti-rotation tab members **40** are in a retracted position within elongated recesses **70** in hanger assembly **10**. It is to be observed that set screws **50** can be tightened in order to “lock” said alignment/anti-rotation tab members **40** in this position. In the configuration depicted in FIG. 4, said alignment/anti-rotation tab members **40** do not extend beyond downwardly-facing landing shoulder **15** of hanger assembly **10**.

(26) FIG. 5 depicts a side perspective view of a hanger assembly **10** of the present invention. Hanger assembly **10** has central through bore **20** extending through said hanger assembly **10**. A plurality of transverse ports **30** extend through hanger assembly **10** from the external surface of said hanger assembly **10** to central through bore **20**. A plurality of circumferential grooves **16** are disposed along the outer surface of hanger assembly **10** for receiving O-rings or other elastomeric sealing members (not depicted in FIG. 5). Elongated recess **70** is disposed along the exterior surface of hanger assembly **10**, while alignment/anti-rotation tab member **40** is moveably disposed within said elongated recess **70**.

(27) Said alignment/anti-rotation tab member **40** has elongated slot **42** and transverse bore **41**. Alignment bolt **60** (not visible in FIG. 5) having head **61** and alignment bolt **62** (not visible in FIG. 5) having head **63** are disposed through said elongated slot **42** and threadedly connected to hanger assembly **10**. Set screw **50** is received within transverse bore **41** of alignment/anti-rotation tab member **40**. It is to be observed that said alignment/anti-rotation tab member **40** can move within said elongated recess **70** when said set screw **50** is sufficiently loosened and does not frictionally or mechanically lock said alignment/anti-rotation tab member **40** against movement. Such movement of said alignment/anti-rotation tab member **40** is in the direction of the longitudinal axis of said elongated recess **70**, which is generally parallel to the longitudinal axis of central through bore **20**. When in a retracted position within elongated recess **70**, said alignment/anti-rotation tab member **40** does not extend beyond downwardly-facing landing shoulder **15** of hanger assembly **10**.

(28) FIG. 6 depicts a side perspective and cut away view of a hanger assembly **10**, while FIG. 7

depicts a side perspective view of said hanger assembly **10**. As depicted in FIGS. **6** and **7**, alignment/anti-rotation tab members **40** are in an extended configuration, wherein said alignment/anti-rotation tab members **40** at least partially extend past downwardly-facing support shoulder **15**.

(29) Referring to FIGS. **6** and **7**, hanger assembly **10** has central through bore **20** extending through said hanger assembly **10**. A plurality of transverse ports **30** extend through hanger assembly **10** from the external surface of said hanger assembly **10** to central through bore **20**. A plurality of circumferential grooves **16** are disposed along the outer surface of hanger assembly **10** for receiving O-rings or other elastomeric sealing members.

(30) Elongated recesses **70** are disposed along the exterior surface of hanger assembly **10**, while an alignment/anti-rotation tab member **40** is moveably disposed within each of said elongated recess **70**. As previously discussed, each alignment/anti-rotation tab member **40** has elongated slot **42** and transverse bore **41**. Alignment bolt **60** having head **61**, and alignment bolt **62** having head **63**, are disposed through said elongated slots **42** and threadedly connected to hanger assembly **10** in threaded bores **71**. Set screw **50** is received within each transverse bore **41** of each alignment/anti-rotation tab member **40**.

(31) It is to be observed that said alignment/anti-rotation tab members **40** can move within each elongated recess **70** when said set screws **50** are sufficiently loosened. Such movement of said alignment/anti-rotation tab member **40** is in the direction of the longitudinal axis of said elongated recess **70**. When in extended position within elongated recess **70**, said alignment/anti-rotation tab member **40** extends or protrudes beyond landing shoulder **15** of hanger **10**. It is to be observed that alignment bolt **60** having head **61** disposed through said each elongated slot **42** limits downward travel of each alignment/anti-rotation tab member **40** within an elongated recess **70**.

(32) FIG. **8** depicts a side sectional view of a hanger assembly **10** of the present invention installed within a central through bore **120** of a wellhead **110**, while FIG. **9** depicts a detailed side sectional view of an alignment/anti-rotation tab member **40** of said hanger assembly **10** depicted in FIG. **8**.

(33) Referring to FIG. **8**, hanger assembly **10** having central through bore **20** is received within central through bore **120** of wellhead **110**. Hanger assembly **10** has downwardly-facing landing shoulder **15** that engages against opposing upwardly-facing shoulder **115** which is disposed on the inner surface of central through bore **120** of wellhead **110**. Landing shoulder **15** and opposing shoulder **115** cooperate to support hanger assembly **10** (and any attached pipe string) when said hanger assembly **10** is landed within central through bore **120** of wellhead **110**. Grooves **16** are disposed along the outer surface of hanger assembly **10** for receiving O-rings or other elastomeric sealing members **17**. Said O-rings or other elastomeric sealing members **17** engage against inner surface **121** of central through bore **120** of wellhead **110** to form a fluid pressure seal between hanger assembly **10** and wellhead **110**.

(34) Each alignment/anti-rotation tab member **40** has elongated slot **42** and transverse bore **41**. Alignment bolt **60** having head **61**, and alignment bolt **62** having head **63**, are disposed through said elongated slots **42** and threadedly connected to hanger assembly **10** in threaded bores **71**. Set screw **50** is received within each transverse bore **41** of each alignment/anti-rotation tab member **40**. In the configuration depicted in FIG. **8**, said alignment/anti-rotation tab members **40** are aligned with slots **140** formed along the inner surface of central through bore **120** of wellhead **110**. As depicted in FIG. **8**, in a retracted position and said alignment/anti-rotation tab members **40** do not extend beyond downwardly-facing landing shoulder **15** and are not received within said slots **140**.

(35) Referring to FIG. **9**, hanger assembly **10** has central through bore **20** and downwardly-facing landing shoulder **15**. Elongated recess **70** is disposed along the exterior surface of hanger assembly **10**. Alignment/anti-rotation tab member **40** is moveably disposed within said elongated recess **70**. Alignment/anti-rotation tab member **40** has elongated slot **42** and transverse bore **41**. Alignment bolt **60** having head **61** and alignment bolt **62** having head **63** are disposed through said elongated slot **42** and are partially received within threaded bores **71** in hanger assembly **10**. Set screw **50** is

received within transverse bore **41** of alignment/anti-rotation tab member **40**.

(36) When set screw **50** is sufficiently loosened, alignment/anti-rotation tab member **40** is permitted to slidably move within said elongated recess **70**. In that case, it is to be observed that alignment bolts **60** and **62** cooperate with elongated slot **42** to ensure that any such travel is exclusively linear and in alignment with the longitudinal axis of elongated recess **70**. Further, bolt heads **61** and **63** cooperate with alignment/anti-rotation tab member **40** to ensure that said alignment/anti-rotation tab member **40** remains within said elongated recess **70** and in contact with hanger assembly **10**.

(37) In the embodiment depicted in FIG. **9**, alignment/anti-rotation tab member **40** is in a retracted position within elongated recess **70** in hanger assembly **10**. In this configuration, said alignment/anti-rotation tab member **40** does not extend beyond landing shoulder **15** of hanger **10**. Set screw **50** can be tightened in order to “lock” said alignment/anti-rotation tab member **40** in this position and against slidable movement within elongated recess **70**.

(38) FIG. **10** depicts a side sectional view of a hanger assembly **10** of the present invention installed within a wellhead **110**, while FIG. **11** depicts a detailed side sectional view of an alignment/anti-rotation tab member **40** of said hanger assembly **10**. Referring to FIG. **10**, hanger assembly **10** having central through bore **20** is received within central through bore **120** of wellhead **110**. Hanger assembly **10** has downwardly-facing landing shoulder **15** that engages against opposing shoulder **115** which is disposed on the inner surface of central through bore **120** of wellhead **110**. Landing shoulder **15** and opposing shoulder **115** cooperate to support hanger assembly **10** (and any attached pipe string) when said hanger assembly **10** is landed within central through bore **120** of wellhead **110**.

(39) Grooves **16** are disposed along the outer surface of hanger assembly **10** for receiving O-rings or other elastomeric sealing members **17**. Said O-rings or other elastomeric sealing members **17** engage against inner surface **121** of central through bore **120** of wellhead **110** to form a fluid pressure seal between hanger assembly **10** and wellhead **110**.

(40) Each alignment/anti-rotation tab member **40** is disposed within an elongated recess **70** and has elongated slot **42** and transverse bore **41**. Alignment bolt **60** having head **61**, and alignment bolt **62** having head **63**, are disposed through said elongated slots **42** and threadedly connected to hanger assembly **10** in threaded bores **71**. Set screw **50** is received within each transverse bore **41** of each alignment/anti-rotation tab member **40**. In the configuration depicted in FIG. **10**, said alignment/anti-rotation tab members **40** are aligned with slots **140** formed along the inner surface of central through bore **120** of wellhead **110**. Further, in the configuration depicted in FIG. **10**, alignment/anti-rotation tab members **40** are in an extended position within elongated recesses **70** in hanger assembly **10**; in this position, said alignment/anti-rotation tab members **40** extend beyond landing shoulder **15** and are received within slots **140** in the inner surface of central through bore **120** of wellhead **110**.

(41) Referring to FIG. **11**, elongated recess **70** is disposed along the exterior surface of hanger assembly **10**. Alignment/anti-rotation tab member **40** is moveably disposed within said elongated recess **70**. Alignment/anti-rotation tab member **40** has elongated slot **42** and transverse bore **41**. Alignment bolt **60** having head **61** and alignment bolt **62** having head **63** are disposed through said elongated slot **42** and are partially received within threaded bores **71** in elongated recess **70** of hanger assembly **10**. Set screw **50** is received within transverse bore **41** of alignment/anti-rotation tab member **40**.

(42) In the configuration depicted in FIG. **11**, alignment/anti-rotation tab member **40** is in an extended position within elongated recess **70** in hanger assembly **10**. In this configuration, said alignment/anti-rotation tab member **40** extends beyond downwardly-facing landing shoulder **15** of hanger **10**. Alignment bolts **60**, each having head **61**, limit downward travel of each alignment/anti-rotation tab member **40** within an elongated recess **70**. Set screw **50** can also be selectively tightened in order to “lock” said alignment/anti-rotation tab member **40** in this position. As

previously noted, alignment bolts **60** and **62** cooperate with elongated slot **42** to ensure that travel of each alignment/anti-rotation tab member **40** is exclusively linear and in alignment with the longitudinal axis of elongated recess **70**. Further, bolt heads **61** and **63** cooperate with each alignment/anti-rotation tab member **40** to ensure that said alignment/anti-rotation tab member **40** remains within said elongated recess **70** and in contact against hanger assembly **10**.

(43) FIG. **12** depicts a side perspective cut-away view of a hanger assembly **10** of the present invention deployed within a wellhead **110**. Hanger assembly **10** has central through bore **20** and is received within central through bore **120** of wellhead **110**; hanger assembly **10** has downwardly-facing landing shoulder **15** that engages against opposing upwardly-facing shoulder **115** which is disposed on the inner surface of central through bore **120** of wellhead **110**. Landing shoulder **15** and opposing shoulder **115** cooperate to support hanger assembly **10** (and any attached pipe string) when said hanger assembly **10** is landed within central through bore **120** of wellhead **110**. Although not visible in FIG. **12**, it is to be understood that alignment/anti-rotation tab members **40** are in an extended position within elongated recesses **70** in hanger assembly **10** and extend into slots **140** in wellhead **110**. In the extended position, said alignment/anti-rotation tab members **40** align hanger assembly **10** within wellhead **110**, and prevent rotation of said hanger assembly **10** within said wellhead **110**.

(44) A plurality of transverse ports **30** extend through hanger assembly **10** and are aligned with transverse ports **130** extending through wellhead **110** when hanger assembly **10** is fully received, landed and properly aligned within central through bore **120** of wellhead **110**. Grooves **16** are disposed along the outer surface of hanger assembly **10** for receiving O-rings or other elastomeric sealing members **17**. Said sealing members **17** disposed within grooves **16** can engage against the inner surface **121** of said central through bore **120** of wellhead **110** to form a fluid pressure seal between said hanger assembly **10** and said wellhead **110**.

(45) In operation, referring to FIG. **4**, a hanger assembly **10** is equipped with at least one alignment/anti-rotation tab member **40**. Each such alignment/anti-rotation tab member **40** is movably disposed within an elongated recess **70** formed along the outer surface of said hanger assembly **10**. During storage or transportation of said hanger assembly **10**, each alignment/anti-rotation tab member **40** can be positioned in a retracted configuration, wherein said alignment/anti-rotation tab members **40** do not extend beyond downwardly-facing landing shoulder **15** of hanger assembly **10**. Set screws **50** can be tightened to secure said alignment/anti-rotation tab members **40** against inadvertent movement.

(46) Prior to landing said hanger assembly **10** within the central through bore of a wellhead (such as wellhead **110**), said set screws **50** can be selectively loosened in order to permit said alignment/anti-rotation tab members **40** to move within said elongated recesses **70**. Referring to FIGS. **6** and **7**, said alignment/anti-rotation tab members **40** can be selectively shifted so that they are positioned in an extended configuration, wherein said alignment/anti-rotation tab members **40** extend beyond downwardly-facing landing shoulder **15** of hanger assembly **10**. In this configuration, bolts **60** limit travel of said alignment/anti-rotation tab members **40** within said elongated recesses **70**. Set screws **50** can be selectively tightened to secure said alignment/anti-rotation tab members **40** in said configuration.

(47) Referring to FIG. **10**, when said alignment/anti-rotation tab members **40** are in an extended configuration during landing of hanger assembly **10** within central through bore **120** of wellhead **110**, said extended alignment/anti-rotation tab members **40** can be aligned with slots **140** formed along the inner surface of central through bore **120** of wellhead **110**. In this configuration, said alignment/anti-rotation tab members **40** extend beyond landing shoulder **15** of hanger assembly **10** and are received within said slots **140** in the inner surface of central through bore **120** of wellhead **110**, while downwardly-facing shoulder **15** of hanger assembly **10** is landed on opposing upwardly-facing support shoulder **115** of wellhead **110**. In this configuration, said alignment/anti-rotation tab members **40** cooperate with slots **140** to align said hanger assembly **10** in a desired radial

orientation along the inner surface of said central through bore **120** of said wellhead **110**. Further, said alignment/anti-rotation tab members **40** cooperate with slots **140** to prevent rotation or radial movement of said hanger assembly **10** within said central through bore **120** of wellhead **110** (such as around the longitudinal axis of central through bore **20** of hanger assembly **10**). Further still, said alignment/anti-rotation tab members **40** cooperate with slots **140** to orient and align transverse ports **30** of hanger assembly **10** with transverse ports **130** of wellhead **110**.

(48) In a preferred embodiment, the preset invention comprises a hanger assembly having the ability to align and orient transverse port(s) in said hanger assembly with port(s) in a surrounding wellhead. The present invention increases the ability to produce fluid(s) from the wellbore through said hanger assembly and aligned outlet port(s) in said hanger assembly and surrounding wellhead without flow restriction. The alignment of said port(s) in the hanger assembly and said wellhead reduces the potential for damage to the hanger and wellhead from fluid erosion. The present invention further permits the production of fluids in a timely manner, thereby eliminating the need for unnecessary downtime for repair or replacement, while increasing production and revenues.

(49) The above-described invention has a number of particular features that should preferably be employed in combination, although each is useful separately without departure from the scope of the invention. While the preferred embodiment of the present invention is shown and described herein, it will be understood that the invention may be embodied otherwise than herein specifically illustrated or described, and that certain changes in form and arrangement of parts and the specific manner of practicing the invention may be made within the underlying idea or principles of the invention.

Claims

1. An alignment and anti-rotation apparatus for a hanger and wellhead comprising: a) a hanger having a central through bore, an external surface, a landing shoulder disposed on said external surface, and at least one elongated channel disposed on said external surface of said hanger, wherein said at least one elongated channel is oriented parallel to a longitudinal axis of said central through bore of said hanger; b) at least one tab member disposed within said at least one elongated channel, wherein said tab member is configured to selectively alternate between a first retracted position and a second extended position, wherein said at least one tab member extends beyond said landing shoulder of said hanger in said second extended position; and c) a wellhead having a central through bore defining an inner surface, further comprising an opposing landing shoulder configured to receive said landing shoulder of said hanger, and at least one elongated slot disposed in said inner surface, wherein said at least one elongated slot is oriented parallel to a longitudinal axis of said central through bore of said wellhead.
2. The apparatus of claim 1, wherein said at least one tab member is received within said at least one elongated slot of said wellhead in said second extended position.
3. The apparatus of claim 2, wherein said at least one tab member and said at least one elongated slot of said wellhead cooperate to align said hanger within said wellhead.
4. The apparatus of claim 2, wherein said at least one tab member and said at least one elongated slot of said wellhead cooperate to align said hanger in a desired orientation along the inner surface of said central through bore of said wellhead.
5. The apparatus of claim 2, wherein said at least one tab member and said at least one elongated slot of said wellhead cooperate to prevent rotation of said hanger relative to said wellhead.
6. The apparatus of claim 2, further comprising: a) at least one transverse port extending through said hanger; and b) at least one transverse port extending through said wellhead.
7. The apparatus of claim 6, wherein said at least one tab member and said at least one elongated slot of said wellhead cooperate to align said at least one transverse port extending through said hanger with said at least one transverse port extending through said wellhead.

8. An alignment and anti-rotation apparatus for a hanger and wellhead comprising: a) a hanger having a central through bore, an external surface, a landing shoulder disposed on said external surface, at least one transverse port extending from said external surface to said central through bore, and at least one elongated channel disposed on said external surface, wherein said at least one elongated channel is oriented parallel to the longitudinal axis of said central through bore of said hanger; b) a wellhead having a central through bore defining an inner surface, further comprising an opposing landing shoulder configured to receive said landing shoulder of said hanger, at least one transverse port extending from an external surface of said wellhead to said central through bore of said wellhead, and at least one elongated slot disposed in said inner surface of said wellhead, wherein said at least one elongated slot is oriented parallel to a longitudinal axis of said central through bore of said wellhead; and c) at least one tab member movably disposed within said at least one elongated channel of said hanger, wherein said tab member is configured to selectively alternate between a first retracted position and a second extended position, and wherein said at least one tab member extends beyond said landing shoulder of said hanger and is received within said at least one elongated slot of said wellhead in said second extended position.

9. The apparatus of claim 8, wherein said at least one tab member and said at least one elongated slot of said wellhead cooperate to align said at least one transverse port of said hanger within said at least one transverse port of said wellhead.

10. The apparatus of claim 8, wherein said at least one tab member and said at least one elongated slot of said wellhead cooperate to prevent rotation of said hanger relative to said wellhead.

11. A method for aligning a hanger within a wellbore comprising: a) providing a hanger having a central through bore, an external surface, a landing shoulder disposed on said external surface, at least one elongated channel disposed on said external surface, and at least one tab member disposed within said at least one elongated channel, wherein said at least one elongated channel is oriented parallel to a longitudinal axis of said central through bore of said hanger, and said at least one tab member is configured to selectively alternate between a first retracted position and a second extended position; b) providing a wellhead having a central through bore defining an inner surface, further comprising an opposing landing shoulder and at least one elongated slot disposed in said inner surface of said central through bore, wherein said at least one elongated slot is oriented parallel to a longitudinal axis of said central through bore of said wellhead; c) selectively shifting said at least one tab member from said first retracted position to said second extended position, wherein said at least one tab member extends beyond said landing shoulder of said hanger in said second extended position; d) landing said hanger within said central through bore of said wellhead; and e) receiving said at least one tab member within said at least one elongated slot of said wellhead.

12. The method of claim 11, wherein said at least one tab member and said at least one elongated slot of said wellhead cooperate to align said hanger in a desired orientation along the inner surface of said central through bore of said wellhead.

13. The method of claim 11, wherein said at least one tab member and said at least one elongated slot of said wellhead cooperate to prevent rotation of said hanger relative to said wellhead.

14. The method of 11, further comprising: a) at least one transverse port extending through said hanger; and b) at least one transverse port extending through said wellhead.

15. The method of claim 14, wherein said at least one tab member and said at least one elongated slot of said wellhead cooperate to align said at least one transverse port extending through said hanger with said at least one transverse port extending through said wellhead.
